

ALIAH UNIVERSITY
 END TERM EXAMINATION, AUTUMN 2024
 PAPER: MATUGBS01 (Engineering Mathematics I)
 Full Marks: 80
 Time Allowed: 3 Hours
 All symbols have their usual meanings
 Use different answer script for different group

GROUP - A (40 marks)

Answer any **EIGHT** of the following questions:

$5 \times 8 = 40$

1. Define a bounded sequence of real number? Define a monotone sequence of real number?
 Using definition show that the sequence $\{a_n\}$ where $a_n = \frac{1}{n^2} + 5$ is bounded and monotone both. 1 + 1 + 3

2. Define a convergent and a divergent sequence of real number. Check whether the sequence $\{x_n\}$ is convergent or divergent, where 2 + 3

$$x_n = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)}.$$

3. Show that $\lim_{n \rightarrow \infty} x^{\frac{1}{n}} = 1$, for $x > 0$. 5

4. Define interval of convergence for a power series? Prove the Geometric series $\sum x^n$ is convergent if $-1 < x < 1$, divergent if $x \geq 1$ and is oscillatory if $x \leq -1$. 1 + 4

5. Test the convergence of the series 5

$$\frac{4}{1.3.5} + \frac{6}{3.5.7} + \frac{8}{5.7.9} + \cdots$$

6. State the D'Alembert's Ratio test. Using D'Alembert's Ratio test prove that the series

$$\frac{1^2}{3} + \frac{2^2}{3^2} + \frac{3^2}{3^3} + \frac{4^2}{3^4} + \cdots$$

is convergent.

1 + 4

7. Show that the function 5

$$f(x) = |x| + |x - 1| + |x - 2|$$

continuous at the points $x = 0, 1, 2$.

8. Define Global maximum and Global minimum for a function $f(x)$?
 Prove that the function $f(x) = x^3 - 9x^2 + 24x - 12$ has a global maximum and a global minimum, also find the global maximum and minimum values of $f(x)$. 1 + 4

9. Let $f : [a, b] \rightarrow \mathbb{R}$ be a differentiable function on (a, b) .
- (a) If $f'(x) < 0$ in (a, b) , then by using mean value theorem prove that $f(x)$ is monotonically decreasing in (a, b) . $2\frac{1}{2}$
- (b) If $f'(x) > 0$ in (a, b) , then by using mean value theorem prove that $f(x)$ is monotonically increasing in (a, b) . $2\frac{1}{2}$
10. State the Taylor's theorem. Expand $\cos x$ in a finite series in Lagrange's form of remainder. $2 + 3$
11. Using L' Hospital rule evaluate $\lim_{x \rightarrow \infty} \left(\frac{\sin x}{x}\right)^{\frac{1}{x}}$. 5
12. If m is a positive integer then show that 5

$$J(m, n) = \frac{(m-1)!}{n(n+1)(n+2) \cdots (n+m-1)}.$$

GROUP - B (40 marks)

Answer any **EIGHT** of the following question:

1. (a) If a line makes an angle $\frac{\pi}{3}$ with x -axis, $\frac{\pi}{4}$ with y -axis and $\theta \in (0, \pi)$ with z -axis, then find the value of θ . $5 \times 8 = 40$
- (b) Convert the point $(-7, -7\sqrt{3})$ to the polar coordinate. 3
2. Find the distance between the lines 2
- $$\frac{x-0}{1} = \frac{y-1}{0} = \frac{z-1}{1} \quad \text{and} \quad \frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$$
3. Find the equation of the plane passing through the point $(1, -1, 2)$ and perpendicular to the planes $2x + 3y - 2z = 5$ and $x + 2y - 3z = 8$. 5
4. Show that the four points $(0, -1, -1)$, $(4, 5, 1)$, $(3, 9, 4)$ and $(-4, 4, 4)$ lie on a plane. Also write down the equation of the plane. 5
5. Find the nature of the quadric surfaces and hence evaluate its volume.
- (a) $3x^2 + 5y^2 + 3z^2 - 12x + 20y - 6z = 5$. 3
- (b) $3x^2 + 3y^2 + 3z^2 - 6x - 6y + 6z = 18$. 2
6. Find the curvature of the curve 5
- $$F(t) = (3 \cos t, 2t, 3 \sin t).$$
7. Find the torsion of the curve 5
- $$F(t) = (t^2, 2t, t^3).$$

8. Evaluate

$$\iint dxdy$$

over the region bounded by $y = x^2$ and $x = y^2$.

5

9. Evaluate

$$\iint (x^2 + y^2) dxdy$$

over the region bounded by $xy = 1$, $y = 0$, $y = x$ and $x = 2$.

5

10. Evaluate

$$\iiint (x + y + z + 1) dxdydz$$

over the region defined by $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq 1$.

5

11. Use Green's Theorem to evaluate

$$\oint_C (xydx + x^2y^3dy)$$

where C is the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 2)$ with positive orientation.

5

12. Use Divergence Theorem to evaluate

$$\iint_S \vec{F} \cdot d\vec{S}$$

where $\vec{F} = \sin(\pi x)\hat{i} + xy^3\hat{j} + (z^2 + 4x)\hat{k}$ and S is the surface of the box with $-1 \leq x \leq 2$, $0 \leq y \leq 1$ and $1 \leq z \leq 4$.

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